

Kaggle Runbook: Pilot Run

Document 07

Status	Ready to use
Version	1.1 · 2026-10-02 23:03 · change log
Purpose	Run the pilot (alignment + zero-shot Whisper) for every recording on a Kaggle GPU, unattended
Related	02 TRD §4.3 , 06 Implementation Plan (Pilot) , Style Guide

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1 What the run does

One notebook cell clones this repository and runs `scripts/kaggle_pilot.sh`, which:

1. installs the dependencies;
2. prints the GPU and PyTorch versions;
3. runs the 24 pilot tests and stops if any fail;
4. checks that Whisper can use the GPU, falling back to CPU if it cannot;
5. pairs every audio file with the DOCX of the same name under `/kaggle/input`, and for each pair runs the alignment (TRD §4.3), segmentation and zero-shot Whisper (small and large-v3-turbo; Swahili, English and auto-detect);
6. writes `summary.md` across recordings plus a `report.md` per recording, and zips them into `/kaggle/working/pilot_reports.zip`.

Your hands-on time is about 10 minutes the first time and 2 minutes after that. The run itself takes roughly 45–90 minutes per two hours of audio on a T4 (an estimate, not yet timed on Kaggle) and uses 1–2 of the approximately 30 weekly GPU hours.

2 One-time setup

Step 1. Verify your phone number

Kaggle only allows GPU and Internet after phone verification.

- kaggle.com → your profile picture → **Settings** → **Phone verification** → verify.
- Skip this if you have done it before.

3 Put the data on Kaggle

Step 2. Create a private dataset

- kaggle.com → **Create** (the + button, top left) → **New Dataset**.
- Drag in the audio and its transcript, e.g. NRCCW_KSM09.MP3 and NRCCW_KSM09.docx. Other recordings can go in the same dataset, each with its own DOCX.
- **Names must match**, ignoring capital letters, spaces, underscores, hyphens and dots. NRCCW_KSM09.MP3 pairs with NRCCW_KSM09.docx.
- Title: kct-audio (any name works).
- Visibility: **Private**. Check this: the recordings are research data.
- Click **Create** and wait until the upload finishes.

4 Create the notebook

Step 3. New notebook

kaggle.com → **Create** → **New Notebook**.

Step 4. Attach the data and switch on GPU and Internet

- Right-hand panel → **Add Input** (or + **Add Data**) → **Your Datasets** → kct-audio → **Add**.
- Right-hand panel → **Session options** (or **Settings**): **Accelerator** = GPU T4 x2 (or GPU P100); **Internet** = On.
- If these options are greyed out, go back to Step 1.

Step 5. Paste the code

Delete any starter code in the first cell and paste exactly:

```
!rm -rf transcribe && git clone -q --depth 1 -b matt/relaxed-meitner-2lfrx6 https://github.com/
↳ mattobryan/transcribe.git
!cd transcribe && bash scripts/kaggle_pilot.sh
```

Optionally rename the notebook to transcribe-pilot by clicking its title.

5 Run it in the background

Step 6. Commit

- Top right → **Save Version** → choose **Save & Run All (Commit)** → **Save**.

- You can close the tab. The run continues on Kaggle’s servers.

Step 7 (optional). Check progress

- Your notebook → **Versions** (clock icon) → the running version → **Logs**.
- The log moves through == [1/5] Installing to == [5/5] Pilot ...
- Near the start, look for 24 passed (tests OK) and Whisper device: cuda (GPU OK).

6 Collect the results

Step 8. Download and send

- When the version shows **Complete**: open it → **Output** tab → download pilot_reports.zip.
- Unzip it and send pilot/summary.md and each pilot/<recording>/report.md.
- The zip holds text and numbers only (reports, alignment and segment lists), no audio.

7 Later runs

- **New recordings**: open the dataset → **New Version** → add the files → **Create**; then open the notebook → **Save Version** → **Save & Run All**. Every run clones the branch afresh, so it always has the latest fixes.
- **Faster or narrower runs**: replace the second code line with one of these.

Goal	Second line
Quick check (about 15 min)	<code>!cd transcribe && MODELS=small LANGS=sw bash scripts/kaggle_pilot.sh</code>
One recording only	<code>!cd transcribe && ONLY=NRCCW_KSM09 bash scripts/kaggle_pilot.sh</code>
Sample of chunks	<code>!cd transcribe && MAX_SEGMENTS=200 bash scripts/kaggle_pilot.sh</code>

8 Troubleshooting

What you see	What to do
GPU or Internet options greyed out	Verify your phone (Step 1).
Could not resolve host github.com	Internet is off: switch it on (Step 4) and commit again.
Found 0 audio/transcript pairs	The dataset is not attached (Step 4), or the audio and DOCX names do not match.
transcript without audio / audio without transcript	A name mismatch for that file; rename it in a new dataset version.
Whisper device: cpu	The run still works, just slower. Send the log lines just above it.
A test fails, or a red error	Download the log (Logs tab, or kaggle_run.log in the output) and send the last 40 lines.
The version fails with no log	Kaggle’s 12-hour or weekly quota limit: rerun with <code>MODELS=small</code> .

9 Using the transcription app online

Changed in v1.1: see the change log.

For an urgent job on a machine without a strong CPU, run the transcription app itself on a free cloud GPU. The notebook is `notebooks/online_app.ipynb`.

1. **Google Colab:** Runtime → Change runtime type → **T4 GPU**. Run the install cell, then cell 2A; click the link it shows. No Kaggle dataset is needed: you upload the audio on the page.
2. **Kaggle:** Accelerator GPU and Internet on. Run the install cell, then cell 2B; open the link it prints (it contains a private token; keep it private and keep the cell running).
3. On the page: **Load audio, Transcribe**, wait, then correct chunk by chunk (Esc, F1–F4).
4. **Download TXT / SRT / JSON** from the top right before closing; the notebook's disk is wiped when the session ends.

Only the text, no page: upload the audio in the notebook's file panel and run `python -m transcribe.app.batch audio.mp3 --out results --language sw`; download `results/audio.txt` and `.srt`.

Limits. Free sessions disconnect when idle and have weekly GPU limits. A recording is processed in one go at roughly several times real time on a T4 (not yet timed on a real recording). The Kaggle link uses a Cloudflare quick tunnel, which could not be tested from the build machine; if it fails, use Colab.

Change log

Times are East Africa Time (EAT, UTC+3). Each entry names the affected section, quotes what it said before and summarises what it says now, and links to the previous text: on GitHub at the exact commit and lines, and as an archived PDF of the previous version.

Version	Date (EAT)	Commit	Summary
1.0	2026-09-24 22:26	c28ea1d	First version: step-by-step Kaggle run of the pilot.
1.1	2026-10-02 23:03	this commit	New section: using the transcription app online (Colab / Kaggle) and the text-only command.

v1.1 • §7 Using the transcription app online (new)

Previous text: none (new section) • [v1.0 PDF](#)

Before. Only the unattended pilot run existed.

Now. How to run the interactive app, or a plain text-only transcription, on a free Colab or Kaggle GPU.